

Technical Design Documentation

1. Project Overview

The Bit&Beam project is an intelligent document management system designed for building-related data. It aims to streamline document workflows using automated classification, metadata extraction, and intelligent search capabilities. The system supports multi-tenancy, ensuring data segregation and access control based on organizations.

2. Architecture

The project follows a modular structure, consisting of frontend, backend, and several services for specific tasks.

3. Key Components

- Frontend: Angular-based UI for user interaction.
 - Located in the [frontend/](#) directory.
 - Uses TypeScript and Angular CLI.
- Backend: C# ASP.NET Core Web API.
 - Located in the [backend/](#) directory.
 - Handles API requests, business logic, and data persistence.
- Database: PostgreSQL with pgai extension.
 - Configuration and schema in [database/](#) directory.
 - Search: Opensearch for indexing and full-text search.
- AI/Extraction:
 - Ollama for AI integration. Located in the [ollama/](#) directory.
 - Apache Tika for metadata extraction. See [TikaService.cs](#).

4. Data Model

The data model is defined using C# classes in the [backend/BUILD.ING/Models/](#) directory. Key entities include:

- User: System users information.
- Building: Stores building details.
- Document: Stores document metadata and file path.
- Organization: Represents a tenant.

5. Key Technologies

Programming Languages: TypeScript, C#, Python

Frameworks/Libraries: Angular, ASP.NET Core

Databases: PostgreSQL

AI systems: Ollama, Apache Tika

Containerization: Docker, Docker Compose

6. Development Guidelines

See [Coding_Guidelines.md](#)

- Protected main and dev branches.
- Feature branches.
- Pull requests require review from at least two other developers.

7. Build and Deployment

Docker Compose is used for orchestrating services. See [docker-compose.yml](#) and [docker-compose-prod.yml](#)

GitHub Actions workflows for linting and CI/CD. See [.github/workflows/](#)